

$$L = E[l(x)] = P(x=2)l(2) + \dots + P(x=n)l(n)$$

$$= \frac{1}{3} \times 2 + \frac{1}{6} \times 3 + \frac{1}{9} \times 3 + \frac{1}{9} \times 3 + \frac{1}{9} \times 3 + \frac{2}{27} \times 3 + \frac{1}{18} \times 4 + \frac{1}{27} \times 4$$

$$\frac{1}{3} \log_2(3) + \frac{1}{6} \log_2(6) + \frac{1}{9} \log_2(9) + \frac{1}{9} \log_2(9) + \frac{1}{9} \log_2(9) + \frac{2}{27} \log_2\left(\frac{27}{2}\right) + \frac{1}{18} \log_2(18) + \frac{1}{27} \log_2(27)$$

$$\frac{2.7017 - 2.759}{2.759} \times 100 \% \quad \infty$$

$$l_i = \left\lceil \log_2 \frac{1}{p_i} \right\rceil \quad p_1 \left\lceil \log_2 \frac{1}{p_1} \right\rceil + p_2 \left\lceil \log_2 \frac{1}{p_2} \right\rceil + \dots + p_n \left\lceil \log_2 \frac{1}{p_n} \right\rceil$$

$$p_i = 2^{-n_i} \quad \log_2 \frac{1}{p_i} = \log_2 2^{n_i} = n_i \quad \text{è un intero}$$

$$\log A = \log \frac{A \cap C}{C} = \log(A \cap C) - \log(C) = \log(A \cap C) + \log\left(\frac{1}{C}\right)$$

$$\sum_i p_i \log A_i = \sum_i p_i \left[\log(A_i \cap C) + \log\left(\frac{1}{C}\right) \right] = \left\{ \sum_i p_i \log(A_i \cap C) \right\} + \log\left(\frac{1}{C}\right) \sum_{i=1}^n p_i$$

$$\frac{P(x) \cap C}{2^{-n(x)}} = \frac{P(x)}{2^{-n(x)}} = \frac{P(x)}{2^{-n(x)}}$$

$$\sum P(x) \log_2(\dots) = \sum P(x) \log_2 \frac{P(x)}{r(x)} = D(P(x), r(x))$$

$$\{x_1, x_2, \dots, x_n\}$$

$$\{p_1, p_2, \dots, p_n\} \dots \dots p_2 \log \frac{1}{p_1} + p_1 \log \frac{1}{p_2} + \dots + p_n \log \frac{1}{p_n}$$

$$P[X=x_i, Y=y_j] = p_i q_j$$

$$H(X, Y) =$$

$$\sum_{i,j} P[X=x_i, Y=y_j] \log_2 \frac{1}{P[X=x_i, Y=y_j]} = \sum_i \sum_j p_i p_j \log_2 \frac{1}{p_i p_j}$$

$$= \sum_i \sum_j p_i p_j \left[\log_2\left(\frac{1}{p_i}\right) + \log_2\left(\frac{1}{p_j}\right) \right]$$

$$= \sum_{i,j} p_i p_j \log \frac{1}{p_i} + \sum_{i,j} p_i p_j \log \frac{1}{p_j}$$

$$= \sum_{i,j} P_i P_j \log \frac{1}{P_i} + \sum_{i,j} \log \left(\frac{1}{P_j} \right) (P_i P_j)$$

$$= \underbrace{\left(\sum_i P_i \log \frac{1}{P_i} \right)}_{H(X)} \underbrace{\left(\sum_j P_j \right)}_{=1} + \underbrace{\left(\sum_j P_j \log \frac{1}{P_j} \right)}_{H(Y)} \underbrace{\left(\sum_i P_i \right)}_{=1} = 2 H(X)$$

$$\sum_i P_j \log \frac{1}{P_j} \left\{ \sum_i P_i \right\}$$

x_1	w_1
x_2	w_2
$\vdots$	$\vdots$
x_n	w_n

x_1, y_1	w_{11}
x_1, y_2	w_{12}
$\vdots$	$\vdots$
x_1, y_n	w_{1n}
x_2, y_1	w_{21}
$\vdots$	$\vdots$
x_2, y_n	w_{2n}
$\vdots$	$\vdots$
x_n, y_1	w_{n1}
$\vdots$	$\vdots$
x_n, y_n	w_{nn}

Exd.
for

x_4, y_1	x_3, y_2	x_3, y_4	x_1, y_4
x_4, x_1	x_3, x_2	x_3, x_4	x_1, x_4
w_{41}	w_{32}	w_{34}	w_{14}
w_{41}	w_{32}	w_{34}	w_{14}

$$H^2 = 2^{2k}$$

$$\underline{01} \quad \underline{10} \quad \underline{01} \quad \underline{01} \quad \underline{01} \quad \underline{01}$$

$$k=1$$

$$b_{2i} = \overline{b_{2i-1}}$$

$$\begin{array}{ll} "0" & \frac{1}{2} \\ "1" & \frac{1}{2} \end{array}$$

x_1	00	→	0	w_1
x_2	01	→	1/2	$w_2 = 0$
x_3	10	→	1/2	$w_3 = 1$
x_4	11	→	00	w_4

$$P_i = P[X=x_i]$$

$$Q_i = Q(Y=y_i)$$

$$E(L(X)) = \sum_i P_i L_i$$

$$\left(\frac{1}{2} \right)^i \left(\frac{1}{2} \right)^{n-i}$$

$$2^N$$

$$\binom{N}{k} 2^{\frac{N}{2}}$$

$$P[X=x_i, Y=y_j] = P[X=x_i] P[Y=y_j] = P_i Q_j$$

$$P[X=x_i, Y=y_j] = P[X=x_i] P[Y=y_j | X=x_i] = P_i Q_j^{(i)}$$

$$= P[Y=y_j] P[X=x_i | Y=y_j] = Q_j P_i^{(j)}$$

$$H(Y/X) = \sum_{i,j} P[X=x_i, Y=y_j] \log \frac{1}{P_i Q_j^{(i)}}$$

$$\sum_{i=1}^M P_i^{(j)} \log_2 \frac{1}{P_i^{(j)}} = H[X | Y = y_j]$$

$$H(Y/X) = \sum_{i,j} \underbrace{P[X=x_i, Y=y_j]}_{P[X=x_i, Y=y_j]} \log_2 \frac{1}{q_j^{(i)}}$$

$$\sum_{j=1}^N q_j H[X | Y = y_j] = \sum_{j=1}^N q_j \sum_{i=1}^M P_i^{(j)} \log_2 \frac{1}{P_i^{(j)}} = \sum_{i,j} \underbrace{(q_j P_i^{(j)})}_{P[X=x_i, Y=y_j]} \log_2 \frac{1}{P_i^{(j)}} = H(X|Y)$$

$$H(X, Y) = \sum_{i,j} P[X=x_i, Y=y_j] \log_2 \frac{1}{P[X=x_i, Y=y_j]} = \sum_{i,j} q_j P_i^{(j)} \log_2 \frac{1}{q_j P_i^{(j)}}$$

$$= \underbrace{\sum_{i,j} q_j P_i^{(j)} \log_2 \frac{1}{P_i^{(j)}}}_{H(X|Y)} + \underbrace{\sum_{i,j} q_j P_i^{(j)} \log_2 \frac{1}{q_j}}_{\sum_j \left\{ q_j \log_2 \frac{1}{q_j} \left[\sum_i P_i^{(j)} \right] \right\} = \sum_j q_j \log_2 \frac{1}{q_j} = H(Y)}$$

$$H(X | Y = y_j) = H(X | Y = y_j) = \sum_i P_i^{(j)} \log_2 \frac{1}{P_i^{(j)}}$$

$$H(X|Y) = \sum_j q_j H(X | Y = y_j)$$

